Assignment 2. Implement the Single-branch network by transforming the baseline method. Write a two page report with results and analysis.

Baseline network. Fusion and orthogonal project

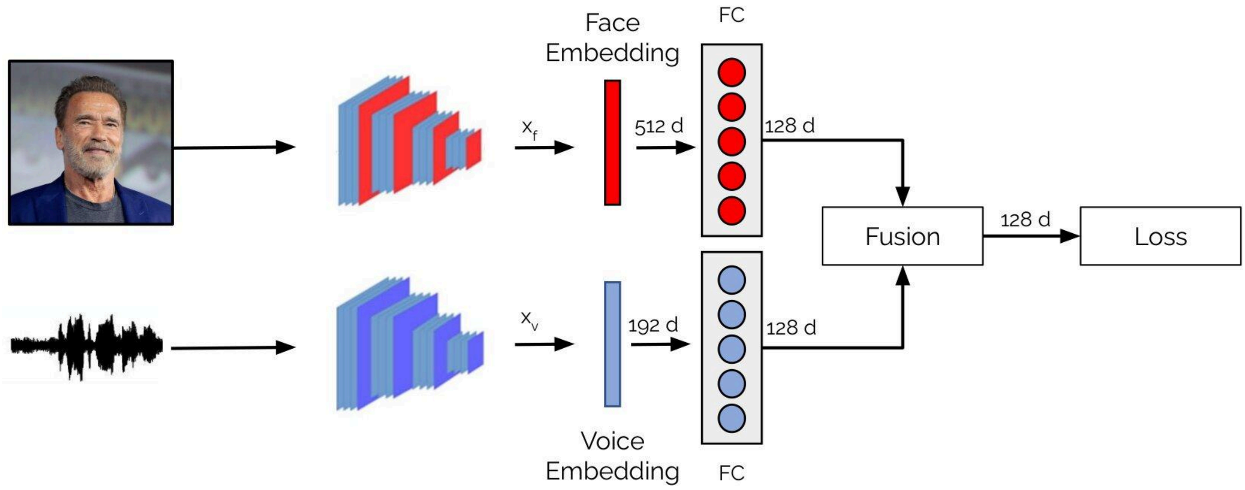

Figure 1. General structure of the baseline network. Face (X_f) and voice (X_v) embeddings are extracted using face and voice encoders, respectively. The extracted embeddings are then passed through linear projection layers to obtain feature representations of dimension ddd. These projected embeddings are subsequently combined using a fusion mechanism to produce multimodal representations, which are used to compute the orthogonal projection loss.

SBNet network**.** Single-branch network

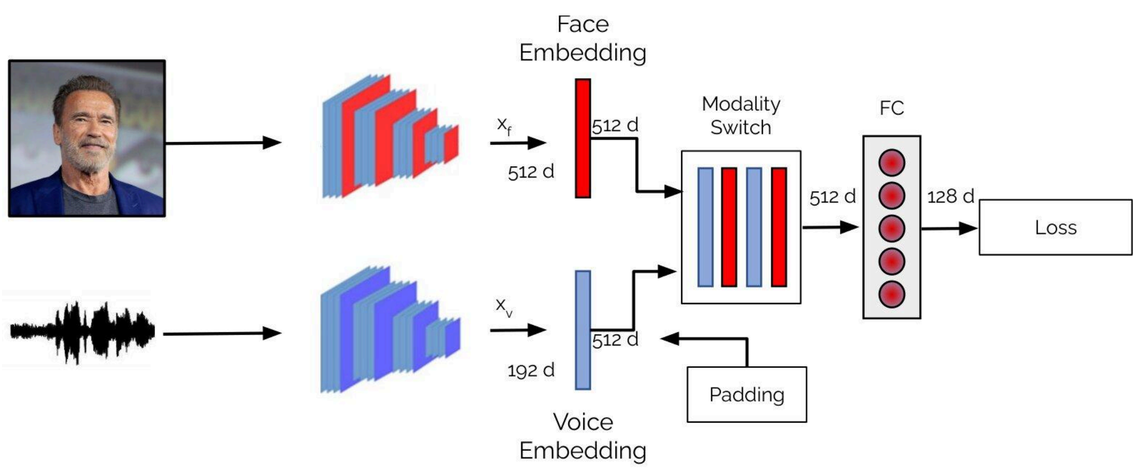

Figure 2. General structure of the single-branch network. Face (X_f) and voice (X_v) embeddings are extracted using face and voice encoders, respectively. In contrast to the baseline, the single-branch network employs a unified branch to learn shared representations. This is achieved by transforming both embeddings to the same embedding size. A modality switch is then applied to randomly sampled and passed through a linear projection layer to compute the orthogonal projection loss.

Training. During training, the single-branch network processes different modalities in a sequential manner. Specifically, the audio embedding of a given sample is first input to the network, followed by the corresponding face embedding of the same sample. This will enable the modal to share the weights across modalities.

Testing. At test time, if complete modalities are present, late fusion is employed by taking the average of the logits obtained from the softmax layer over all modalities. In the case of only one modality, the fusion mechanism is not employed.

The single-branch network learns inter-modality representations across modalities, bringing them into a similar latent space. This results in a richer representation that is complemented by both modalities and yields a robust model when one of the modalities is missing. In other words, when a modality is missing, the available modality enables the use of the multimodal knowledge from the shared inter-modality representation.

Background. We have already implemented the single-branch network to establish face-voice association by leveraging the cross-modal verification task. Here is the Github repository: https://github.com/msaadsaeed/SBNet/tree/master/ssnet_fop

You can change the code to perform the classification results.

Updated baseline results. I have provided the updated test.py script of the baseline network.

Config.	**P3 (Face-Audio Eng.)**	**P4 (Audio Eng.)**	**P5 (Face-Audio Ger)**	**P6 (Audio Ger)**	**Avg.**
Baseline Network					
Face-Audio (Eng.)	78.45	63.53	73.39	63.43	69.70
Face-Audio (Ger.)	89.77	69.30	90.95	62.07	78.02
Single-branch Network					
Face-Audio (Eng.)	-	-	-	-	-
Face-Audio (Ger.)	-	-	-	-	-